Milestone M3 (Lane B): Generalized Momentum Observer Formulation, Implementation & Validation Directive

Project IIT/SRIC/R/AEH/2026/104 • Audited Release v2.0.0-PRELIMINARY • Status: PRELIMINARY (Pending Frozen MS02P05 Simulink Sweep Signals)

Reproducibility Field	Verified Specification & Archive Provenance	Repository Source & Checksum
Code & Report Version	`m3_observer_suite.m` (`v2.0.0-PRELIMINARY`) • Output Archive: `M3_observer_audit_results.mat`	M2_README.md, CODE_EXPLAINABILITY_PROTOCOL.md
Ingested Null Archive	M2_null.mat (N_{raw} = 3,001 @ 100.0 Hz, $t \in [0, 30]$ s, `TrimStart = 5.0 s`, N_{30} = 901, N_{eval} = 751)	SHA-256: `74a2ed3fab4c1f39...` Logged: `2026-08-25 18:57:29` (git 5b3861b, R2025b)
Ingested Load Archive	M2_100kN_0p1111Hz_sp1p50.mat (F_y = 100 kN $sin(2\pi t/9)$ at bail station $s = d_4 + 1.50$ m)	SHA-256: `6050507d22c8e764...` Logged: `2026-08-25 18:57:30` (git 5b3861b, R2025b)
Locked M1 Plant & Observer Config	M_d = 54,300 kg, I_{zz} = 287,900 kg·m^2, c = −1.32 m, f_{v3} = 3,800, f_{v4} = 1,473, f_{c3} = 340 N·m, f_{c4} = 480 N • Frozen Gains: $k_3 = k_4$ = 20.0 s^{-1} (M = 67, $\Delta\tau$ = 0.498 ms) • Portable RNG Seed: `202603`	shovel_params.m, equations_master.tex, recover_load.m
Validation Status	**PRELIMINARY** — All metrics are generated from the verified pre-P05 Simscape archives above and remain **PRELIMINARY** until re-run on frozen `MS02P05` (`05_configuration_sweep`) Simulink signals	AI_COLLABORATION_PROTOCOL.md, research_log.md

Foundational Engineering Premise & Audit Summary: A 30 Hz observer (T_s = 1/30 s ≈ 33.33 ms) does not need to resolve the internal micro-velocity $tanh(\dot{q}/\varepsilon)$ transition (ε_3 = 3×10^{-4} rad/s, ε_4 = 10^{-3} m/s) between samples to estimate external excavation loads reliably. Sub-stepping the observer ODE at 1.0 ms (M = 33) or 0.5 ms (M = 67) eliminates numerical integration truncation error relative to the assumed linear First-Order Hold (FOH) inter-sample trajectory—reducing discretization RMSE against the exact analytical FOH solution on M2_100kN_0p1111Hz_sp1p50.mat from 560.39 N·m (M = 1 macro Euler) to 7.26 N·m (0.5 ms Euler) and 1.20×10^{-7} N·m (0.5 ms RK4). However, sub-stepping does *not* create unmeasured velocity information between 30 Hz samples: physical load-tracking RMSE against the true generalized load Q_{pred} plateaus at exactly 16.69 kN·m (q_3, 2.27% of peak) and 1.77 kN (d_4, 2.04% of peak) for all $M \geq 33$ at the frozen gain $k_3 = k_4$ = 20 s^{-1}.

1. Mathematical Derivation, Sign Convention & Archive Ingestion Audit

Both observer branches share the Stage 1 (M1) Euler–Lagrange plant equations documented in equations_master.tex and shovel_params.m. For the generalized coordinate vector $q = [q_3, d_4]^T$ (q_3 in rad counter-clockwise from horizontal, d_4 in m), the radial distance from the saddle pivot to the composite dipper–payload center of gravity (COG) is:

$$r(d_4) = d_4 + x_{ref} - L_{COG} = d_4 + c, \quad \text{where } c = 1.50\ \text{m} - 2.82\ \text{m} = -1.32\ \text{m}.$$

The unified equation of motion under an unknown external generalized excavation load $\tau_L = [\tau_{L,3}, F_{L,4}]^T$ and measured actuator effort $\tau_a = [\tau_{a,3}, F_{a,4}]^T$ is:

$$D(q)\,\ddot{q} + C(q,\dot{q})\,\dot{q} + G(q) + F_v\,\dot{q} + F_c(\dot{q}) = \tau_a + \tau_L$$

$$D(q) = \text{diag}(I_{zz} + M_d\,r^2,\ M_d), \quad G(q) = [\,M_d\,g\,r\cos(q_3),\ M_d\,g\sin(q_3)\,]^T, \quad F_v\,\dot{q} = [\,f_{v3}\,\dot{q}_3,\ f_{v4}\,\dot{d}_4\,]^T,$$
$$F_c(\dot{q}) = [\,f_{c3}\tanh(\dot{q}_3/\varepsilon_3),\ f_{c4}\tanh(\dot{d}_4/\varepsilon_4)\,]^T.$$

1.1 Verification of Christoffel Matrix C(q, q̇) and Identity Ḋ(q) = C(q, q̇) + C(q, q̇)ᵀ

Factoring $C(q,\dot{q})$ via Christoffel symbols of the first kind $C_{ij} = \tfrac{1}{2}\sum_k (\partial D_{ij}/\partial q_k + \partial D_{ik}/\partial q_j - \partial D_{jk}/\partial q_i)\,\dot{q}_k$ with $\partial D_{11}/\partial d_4 = 2\,M_d\,r$ yields $C_{11} = M_d\,r\,\dot{d}_4$, $C_{12} = M_d\,r\,\dot{q}_3$, $C_{21} = -M_d\,r\,\dot{q}_3$, and $C_{22} = 0$, matching Eq. (8) of equations_master.tex. Differentiating $D(q)$ confirms:

$$\dot{D}(q) = \text{diag}(2\,M_d\,r\,\dot{d}_4,\ \ 0) = C(q,\dot{q}) + C(q,\dot{q})^T, \quad \dot{D}(q) - 2C(q,\dot{q}) = [\,0,\ \ -2\,M_d\,r\,\dot{q}_3\,;\ \ +2\,M_d\,r\,\dot{q}_3,\ \ 0\,] = -(\dot{D} - 2C)^T,$$

$$C(q,\dot{q})^T\,\dot{q} = [\,(M_d\,r\,\dot{d}_4)\,\dot{q}_3 - (M_d\,r\,\dot{q}_3)\,\dot{d}_4,\ \ (M_d\,r\,\dot{q}_3)\,\dot{q}_3 + 0\,]^T = [\,0,\ \ +M_d\,r\,\dot{q}_3^{\,2}\,]^T.$$

In Channel 1 (saddle rotation q_3), $\dot{p}_3 = (I_{zz} + M_d\,r^2)\,\ddot{q}_3 + 2\,M_d\,r\,\dot{d}_4\,\dot{q}_3$ absorbs the full Coriolis torque, so $(C^T\dot{q})_3 = 0$. In Channel 2 (crowd translation d_4), $\dot{p}_4 = M_d\,\ddot{d}_4$ contains no centrifugal term, so $+M_d\,r\,\dot{q}_3^{\,2}$ enters $\dot{p}_4$ with a positive sign. Substituting $D(q)\,\ddot{q} = \dot{p} - (C + C^T)\,\dot{q}$ yields the exact generalized momentum rate equation:

$$\dot{p} = \tau_a + \tau_L - G(q) - F_v\,\dot{q} - F_c(\dot{q}) + C(q,\dot{q})^T\,\dot{q}$$

1.2 Audited Sign Reconciliation with Lane A (recover_load.m) & MAT Archive Verification

In recover_load.m (line 34) and M2_nomenclature.md, the plant equation is $D\,\ddot{q} + C\,\dot{q} + G + f = \tau_{actuator} + Q_{ext}$, where $Q_{ext} = J_s(q)^T\,F = [R.sig.Qq3_pred;\ R.sig.Qd4_pred]$ is the generalized external load and $e_\tau = \tau_{measured} - \tau_{model} = -Q_{ext} = [R.sig.ref3;\ R.sig.ref4]$ is the PID actuator residual. Comparing $D\,\ddot{q} + C\,\dot{q} + G + F_v\,\dot{q} + F_c = \tau_a + \tau_L$ with $\tau_a = \tau_{actuator}$ establishes the exact identity:

$$\tau_L \equiv +Q_{ext} = +J_s(q)^T\,F = [\,\text{R.sig.Qq3_pred}\ ;\ \ \text{R.sig.Qd4_pred}\,] = -e_\tau = -[\,\text{R.sig.ref3}\ ;\ \ \text{R.sig.ref4}\,].$$

Archive Validation & Check 0 (`P.model_has_coulomb = false`): Calling `load_and_validate_mat_archive` on stage1_results.mat correctly aborts with `m3_observer_suite:SummaryOnlyMatFile` because `stage1_results.mat` stores only summary identification metrics (`results.ident`) without time-series vectors. Loading the full v7.3 archives M2_null.mat and M2_100kN_0p1111Hz_sp1p50.mat verifies all 20 signals ($N_{raw} = 3{,}001$ at $100.0\ Hz$, $q_3 \in$

$[30°, 60°]$, $d_4 \in [6.7505, 10.7495]\ m$, $r \in [5.4305, 9.4295]\ m$, $s_{bail} \in [8.2505, 12.2495]\ m$, dynamic consistency $7.88\times10^{-7}\ N\cdot m$ after $30\ Hz$ spline resampling). Because shovel_params.m line 50 had `P.model_has_coulomb = false` during the `2026-08-25` Simulink runs ($max|e_{\tau 3} - ref3| = 2.00\times10^{-6}\ N\cdot m$), running the observers on the unmodified raw archive (`Check 0`) gives Branch B RMSE $[53.47\ N\cdot m, 4.36\ N]$ and Branch A RMSE $[378.89\ N\cdot m, 480.72\ N]$. Embedding the M3 regularized Coulomb friction $+F_c(\dot{q})$ into the plant actuator effort $\tau_a = \tau_{meas,slx} + F_c(\dot{q})$ restores the M3 plant equation while preserving the exact logged Simscape kinematics.

Channel / Quantity	Symbol & Algebraic Expression	SI Base Breakdown	Engineering Unit
Channel 1 (Saddle Revolute)	$q_3,\ \dot{q}_3,\ D_{11} = I_{zz} + M_d\, r^2,\ p_3 = D_{11}\, \dot{q}_3$	p_3: $[kg\cdot m^2\cdot s^{-1}]$	$rad,\ rad/s,\ kg\cdot m^2,\ N\cdot m\cdot s$
Saddle Momentum Rate & Output	$\dot{p}_3,\ \tau_{a,3},\ \tau_{L,3},\ M_d\, g\, r\, cos(q_3),\ f_{v3}\, \dot{q}_3,\ f_{c3}\, tanh(\dot{q}_3/\varepsilon_3),\ \hat{\tau}_{L,3} = k_3(p_3 - z_3)$	$[kg\cdot m^2\cdot s^{-2}]$ $((C^T\dot{q})_3 = 0)$	$N\cdot m$ (k_3 in s^{-1})
Channel 2 (Crowd Prismatic)	$d_4,\ \dot{d}_4,\ D_{22} = M_d,\ p_4 = M_d\, \dot{d}_4$	p_4: $[kg\cdot m\cdot s^{-1}]$	$m,\ m/s,\ kg,\ N\cdot s$
Crowd Momentum Rate & Output	$\dot{p}_4,\ F_{a,4},\ F_{L,4},\ M_d\, g\, sin(q_3),\ f_{v4}\, \dot{d}_4,\ f_{c4}\, tanh(\dot{d}_4/\varepsilon_4),\ M_d\, r\, \dot{q}_3^2,\ \hat{F}_{L,4} = k_4(p_4 - z_4)$	$[kg\cdot m\cdot s^{-2}]$	N (k_4 in s^{-1})

2. Continuous-Time & Discrete-Time (30 Hz) Observer Formulations

2.1 Branch A: Friction-Aware Continuous–Discrete Momentum Observer (Primary Branch)

Branch A explicitly includes the regularized Coulomb friction vector $F_c(\dot{q}) = [340\ tanh(\dot{q}_3/\varepsilon_3),\ 480\ tanh(\dot{d}_4/\varepsilon_4)]^T$ inside the known model dynamics vector $h_A(q, \dot{q}, \tau_a)$:

$$h_A(q, \dot{q}, \tau_a) = \tau_a - G(q) - F_v\, \dot{q} + C(q, \dot{q})^T\, \dot{q} - F_c(\dot{q}) \quad \Rightarrow \quad \dot{z}_A(t) = h_A(t) + K_O\,(\,p(t) - z_A(t)\,), \quad \hat{\tau}_{L,A}(t) = K_O\,(\,p(t) - z_A(t)\,).$$

When the plant model is exact, $\dot{\hat{\tau}}_{L,A} = K_O(\tau_L - \hat{\tau}_{L,A})$, so each channel acts as an exact decoupled first-order low-pass filter $\hat{\tau}_{L,A,i}(s) = (k_i / (s + k_i))\, \tau_{L,i}(s)$. At discrete $30\ Hz$ measurement updates $t_k = k\, T_s$ ($T_s = 1/30\ s$), the observer linearly interpolates $p(t_k + \tau) = p_k + (\tau/T_s)(p_{k+1} - p_k)$ and $h_A(t_k + \tau) = h_{A,k} + (\tau/T_s)(h_{A,k+1} - h_{A,k})$ across M uniform sub-steps of size $\Delta\tau = T_s / M$ ($M = 33 \Rightarrow 1.010\ ms$; $M = 67 \Rightarrow 0.498\ ms$):

$$z_A^{(m+1)} = z_A^{(m)} + \Delta\tau\,[\,h_A(t_k + m\Delta\tau) + K_O\,(\,p(t_k + m\Delta\tau) - z_A^{(m)}\,)\,], \quad \hat{\tau}_{L,A,k+1} = K_O\,(\,p_{k+1} - z_{A,k+1}\,),$$

$$z_{A,i,k+1}^{FOH} = e^{-\alpha_i}\, z_{A,i,k} + (1 / k_i)\,[\,(1 - e^{-\alpha_i})\, u_{i,k} + (\,1 - (1 - e^{-\alpha_i})/\alpha_i\,)\,(u_{i,k+1} - u_{i,k})\,], \quad (\alpha_i = k_i\, T_s,\ \ u_k = h_{A,k} + K_O\, p_k).$$

2.2 Branch B: Friction-Uncertainty Observer & Strict Separation of Error Bounds

Branch B omits Coulomb friction from its model vector: $h_B = \tau_a - G(q) - F_v\, \dot{q} + C(q, \dot{q})^T\, \dot{q}$, so $\dot{p} = h_B + (\tau_L - F_c(\dot{q}))$ and $\hat{\tau}_{L,B} = K_O(p - z_B)$. Crucially, the total instantaneous estimation error of Branch B decomposes into three distinct physical terms:

$$\tau_L(t_k) - \hat{\tau}_{L,B}(t_k) = [\,\hat{\tau}_{L,A}(t_k) - \hat{\tau}_{L,B}(t_k)\,] + [\,\tau_{L,filt}(t_k) - \hat{\tau}_{L,A}(t_k)\,] + [\,\tau_L(t_k) - \tau_{L,filt}(t_k)\,].$$

- **Term 1 — Filtered Friction Discrepancy ($\hat{F}_{c,filt} = \hat{\tau}_{L,A} - \hat{\tau}_{L,B}$):** Governed by $\dot{\hat{F}}_{c,filt} + K_O\,\hat{F}_{c,filt} = K_O\,F_c(\dot{q})$. Since $|F_{c,3}| \le 340\ N{\cdot}m$, $|F_{c,4}| \le 480\ N$, and $\int_0^\infty k_i\,e^{-k_i\,t}\,dt = 1$, Young's inequality guarantees $|\hat{\tau}_{L,A,3} - \hat{\tau}_{L,B,3}| \le 340\ N{\cdot}m$ and $|\hat{F}_{L,A,4} - \hat{F}_{L,B,4}| \le 480\ N$ at 100% of samples (`CovFricPct = 100.0%`).
- **Term 2 — FOH Inter-Sample Chord Error ($\tau_{L,filt} - \hat{\tau}_{L,A}$):** Arises because linear interpolation across $T_s = 33.33\ ms$ approximates curved inertial/gravity trajectories ($53.47\ N{\cdot}m$ RMS on q_3, $4.36\ N$ RMS on d_4). Evaluated via lag-compensated coverage (`CovLagPct`: $|\tau_{L,filt,i} - \hat{\tau}_{L,B,i}| \le f_{ci}$).
- **Term 3 — Dynamic Observer Lag ($\tau_L - \tau_{L,filt} \approx \dot{\tau}_L / k_i$):** For rapidly varying loads ($|\dot{Q}_{q3}| \approx 512\ kN{\cdot}m/s$ in Case C), the $50\ ms$ observer lag reaches $25.5\ kN{\cdot}m$, which is 75× larger than $f_{c3} = 340\ N{\cdot}m$. Therefore, **$\pm F_c$ alone never bounds total instantaneous dynamic tracking error** (measured by `CovRawPct`: $|\tau_{L,i} - \hat{\tau}_{L,B,i}| \le f_{ci}$).

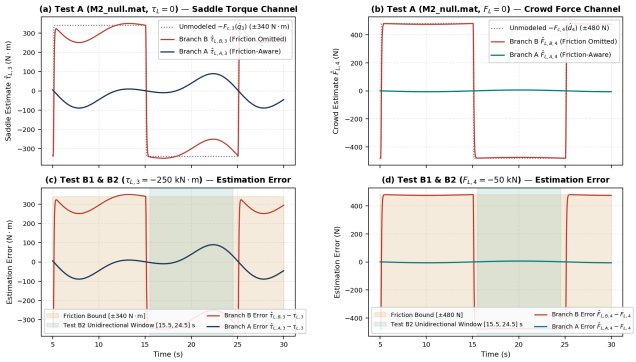


Figure 1: Verification on M2_null.mat under Test A (Zero Load) and Test B1/B2 (Constant Load: −250 kN·m, −50 kN). Under zero load, Branch B misattributes unmodeled Coulomb friction directly as a false load (RMSE 300.43 N·m on q3, 472.84 N on d4), while Branch A cuts false load by 82.2% on q3 (RMSE 53.47 N·m) and 99.1% on d4 (RMSE 4.36 N). Shaded teal window [15.5, 24.5] s highlights the unidirectional segment (Test B2).

---

3. Numerical Integration Convergence Study (1.0 ms vs. 0.5 ms vs. Exact FOH)

Evaluated directly on M2_100kN_0p1111Hz_sp1p50.mat (Case C, $k_3 = k_4 = 20\ s^{-1}$, $N_{eval} = 751$ over $t \in [5.0, 30.0]\ s$), Table 1 and Figure 2 separate numerical ODE integration error against the analytical FOH solution from total physical tracking error against Q_{pred}.

Integration Scheme (T_s = $33.333\ ms$)	Sub-Steps M ($\Delta\tau$)	q_3 Discretization RMSE vs. FOH ($N{\cdot}m$)	d_4 Discretization RMSE vs. FOH (N)	q_3 Total RMSE vs. True Q_{q3} ($N{\cdot}m$)	d_4 Total RMSE vs. True Q_{d4} (N)
Macro Explicit Euler	$M = 1$ ($33.333\ ms$)	560.3862	52.1416	16,363.91	1,756.02
Sub-Stepped Explicit Euler (1.0 ms)	$M = 33$ ($1.010\ ms$)	14.7721	1.3741	16,685.41	1,766.18
Sub-Stepped Explicit Euler (0.5 ms)	$M = 67$ ($0.498\ ms$)	7.2646	0.6757	16,689.89	1,766.32

Integration Scheme (T_s = $33.333\ ms$)	Sub-Steps M ($\Delta\tau$)	q_3 Discretization RMSE vs. FOH ($N{\cdot}m$)	d_4 Discretization RMSE vs. FOH (N)	q_3 Total RMSE vs. True Q_{q3} ($N{\cdot}m$)	d_4 Total RMSE vs. True Q_{d4} (N)
Sub-Stepped Explicit Euler (0.1 ms)	$M = 333$ ($0.100\ ms$)	1.4599	0.1358	16,693.36	1,766.43
Sub-Stepped RK4 (1.0 ms)	$M = 33$ ($1.010\ ms$)	2.0582×10^{-6}	1.9143×10^{-7}	16,694.23	1,766.46
Sub-Stepped RK4 (0.5 ms)	$M = 67$ ($0.498\ ms$)	1.2046×10^{-7}	1.1205×10^{-8}	16,694.23	1,766.46
Exact Analytical FOH	Closed-Form ($M = 1$)	0.0000	0.0000	16,694.23	1,766.46

Key Numerical Takeaway: Halving the explicit Euler sub-step from $1.010\ ms$ ($M = 33$) to $0.498\ ms$ ($M = 67$) cuts the FOH discretization RMSE by $2.033\times$ ($14.77\ N{\cdot}m \rightarrow 7.26\ N{\cdot}m$ on q_3; $1.37\ N \rightarrow 0.68\ N$ on d_4), while RK4 at $0.5\ ms$ converges to $1.20{\times}10^{-7}\ N{\cdot}m$. Yet across all converged schemes ($M \geq 33$), the physical tracking RMSE against Q_{pred} remains invariant at $16{,}694.23\ N{\cdot}m$ and $1{,}766.46\ N$, proving that sub-stepping refines ODE integration of the assumed FOH model without increasing $30\ Hz$ measurement information.

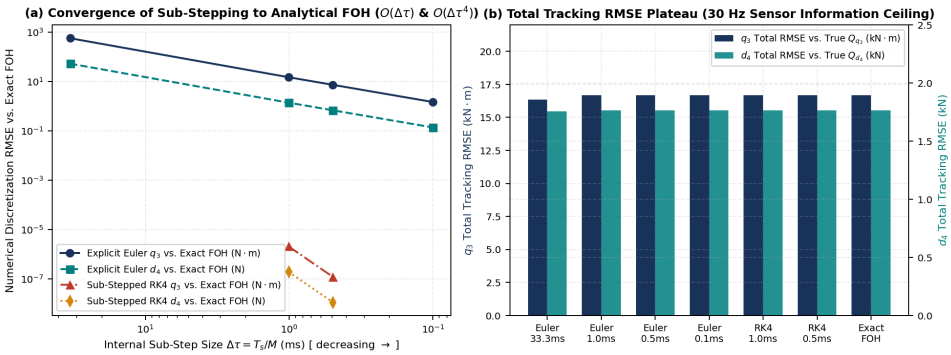


Figure 2: Log-log convergence of numerical integration error $||\hat{\tau}_{L,A}^{(\Delta\tau)} - \hat{\tau}_{L,A}^{FOH}||_{RMSE}$ (left) contrasted with the invariant total load-tracking RMSE against R.sig.Qq3_pred and R.sig.Qd4_pred in M2_100kN_0p1111Hz_sp1p50.mat (right).

4. Observer Gain Selection Study & Pre-M4 Gain Freeze

Sweeping $k \in \{2, 5, 10, 15, 20, 30, 45, 55\}\ s^{-1}$ across M2_100kN_0p1111Hz_sp1p50.mat and M2_null.mat (Table 2, Figure 3) quantifies the trade-off between tracking lag ($\tau_O = 1/k$), $30\ Hz$ FOH chord error amplification on $M2_null.mat$, and synthetic velocity-noise amplification ($k_3\, D_{11}(d_4)\, \delta\dot{q}_3$, where $D_{11} \in [1.89{\times}10^6, 5.12{\times}10^6]\ kg{\cdot}m^2$ for $d_4 \in [6.75, 10.75]\ m$).

Gain k (s^{-1})	τ_O = $1/k$ (ms)	Theory $t_{s,2\%}$ (s)	Macro $k\,T_s$	Case C Sin RMSE (q_3 kN·m / d_4 kN)	Case C Phase Lag (ms / deg)	Case D Step $t_{conv,2\%}$ (s)	Case A Null RMSE / RevMax ($N{\cdot}m$)	Case G Synth. Noise RMSE (q_3 kN·m / d_4 kN)
2.0	500.0	2.000	0.067	159.31 / 16.98	478.6 / 19.14°	1.967	5.38 / 1.43	160.70 / 17.00
5.0	200.0	0.800	0.167	66.28 / 7.04	198.3 / 7.93°	0.767	13.46 / 1.77	84.19 / 7.17
10.0	100.0	0.400	0.333	33.31 / 3.53	99.8 / 3.99°	0.400	26.88 / 3.55	104.63 / 4.30
15.0	66.7	0.267	0.500	22.24 / 2.36	66.6 / 2.67°	0.267	40.23 / 5.33	143.15 / 4.19
20.0 (Frozen)	**50.0**	**0.200**	**0.667**	**16.69 / 1.77**	**50.1 / 2.00°**	**0.200**	**53.47 / 7.10**	**180.05 / 4.74**
30.0	33.3	0.133	1.000	11.16 / 1.18	33.5 / 1.34°	0.133	79.48 / 10.62	243.76 / 6.08
45.0	22.2	0.089	1.500	7.48 / 0.79	22.4 / 0.90°	0.100	116.88 / 15.73	315.37 / 7.74
55.0	18.2	0.073	1.833	6.15 / 0.64	18.4 / 0.74°	0.067	140.44 / 18.99	349.81 / 8.56

Justification for Freezing $k_3 = k_4 = 20.0\ s^{-1}$: Freezing $k_3 = k_4 = 20.0\ s^{-1}$ ($\tau_O = 50.0\ ms$, $t_{conv,2\%} = 0.200\ s = 6$ samples, $k\,T_s = 0.667 < 1$) achieves $2.00°$ phase lag (-0.065% amplitude error) on `M2_100kN_0p1111Hz_sp1p50.mat` while keeping the zero-load FOH chord RMSE on `M2_null.mat` at $53.47\ N{\cdot}m$ (15.7% of f_{c3}) and $4.36\ N$ (0.91% of f_{c4}), whereas pushing $k \to 55\ s^{-1}$ drives the zero-load FOH chord RMSE up to $140.44\ N{\cdot}m$ and nearly doubles synthetic noise amplification.

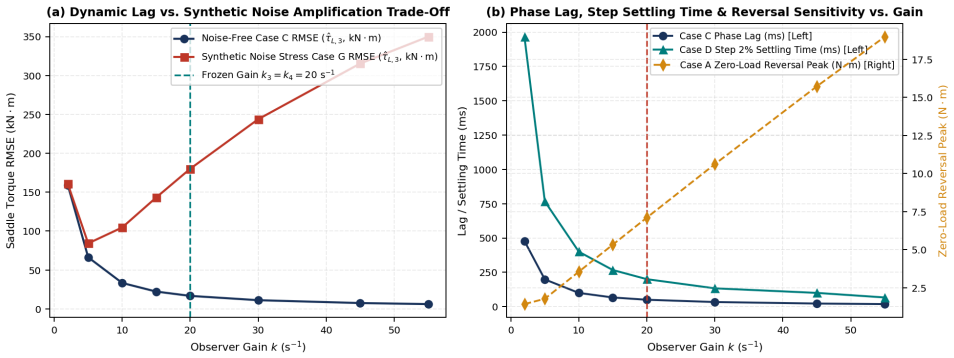


Figure 3: Multi-objective observer gain sweep across k ∈ [2, 55] s^{-1} on M2_100kN_0p1111Hz_sp1p50.mat and M2_null.mat showing the frozen gain $k_3 = k_4 = 20\ s^{-1}$.

5. Audited Comparative Validation Suite (Test Cases A, B1, B2, C, D, E, F, G)

All eight test cases in Tables 3 and 4 are generated directly by `m3_observer_suite.m` (`v2.0.0-PRELIMINARY`) from the ingested M2_null.mat and M2_100kN_0p1111Hz_sp1p50.mat archives ($k_3 = k_4 = 20\ s^{-1}$, $M = 67$, RK4). Three separate coverage columns are reported for every case: **CovFric** ($|\hat{\tau}_{L,A} - \hat{\tau}_{L,B}| \le f_c$), **CovLag** ($|\tau_{L,filt} - \hat{\tau}_L| \le f_c$), and **CovRaw** ($|\tau_L - \hat{\tau}_L| \le f_c$).

5.1 Channel 1 Performance Summary: Saddle Torque $\tau_{L,3}$ (Units: N·m)

Test Case & Archive Provenance	Br.	RMSE ($N{\cdot}m$)	MAE ($N{\cdot}m$)	Bias ($N{\cdot}m$)	Max |Err| ($N{\cdot}m$)	Rev MAE Pre / Dur / Post	CovFric / CovLag / CovRaw (%)
A: Zero Load (`M2_null.mat`, $N=751$)	A B	53.47 300.43	42.44 297.62	−12.60 +51.23	89.24 349.65	9.1 / 2.6 / 11.2 333.3 / 252.5 / 324.4	100.0 / 100.0 / 100.0 100.0 / 75.4 / 75.4
B1: Constant Rev. ($-250\ kN{\cdot}m$, $N=751$)	A B	53.47 300.43	42.44 297.62	−12.60 +51.23	89.24 349.65	9.1 / 2.6 / 11.2 333.3 / 252.5 / 324.4	100.0 / 100.0 / 100.0 100.0 / 75.4 / 75.4
B2: Constant Unidir. ($t \in [15.5, 24.5]\ s$, $N=271$)	A B	51.63 306.35	40.25 304.10	+35.90 −304.10	89.24 349.65	0.0 / 0.0 / 3.2 0.0 / 0.0 / 343.0	100.0 / 100.0 / 100.0 100.0 / 68.3 / 68.3
C: Sinusoidal (`M2_100kN`, $734.2\ kN{\cdot}m$ pk)	A B	16,694.23 16,662.07	14,891.99 14,852.84	−1,642.05 −1,578.21	25,538.98 25,878.98	8,511 / 11,651 / 11,091 8,714 / 11,701 / 10,979	100.0 / 100.0 / 1.1 100.0 / 75.4 / 0.8
D: Step Load ($300\ kN{\cdot}m$ at $t=12\ s$, $N=751$)	A B	9,310.90 9,293.70	641.41 886.15	−611.80 −547.97	218,960.36 218,620.36	9.1 / 2.6 / 11.2 333.3 / 252.5 / 324.4	100.0 / 100.0 / 98.7 100.0 / 75.4 / 76.3
E: Time-Varying Digging ($320\ kN{\cdot}m$ pk)	A B	3,764.74 3,900.88	2,707.39 2,800.37	−30.73 +33.11	9,399.96 9,739.96	3,493 / 837 / 2,052 3,833 / 944 / 2,345	100.0 / 100.0 / 18.6 100.0 / 75.4 / 20.1

Test Case & Archive Provenance	Br.	RMSE ($N \cdot m$)	MAE ($N \cdot m$)	Bias ($N \cdot m$)	Max |Err| ($N \cdot m$)	Rev MAE Pre / Dur / Post	CovFric / CovLag / CovRaw (%)
F: Reversal-Focused (*1.5 kN·m, 0.18 Hz*)	A B	79.92 308.81	67.69 300.08	−16.40 +47.44	173.93 434.11	73.3 / 59.0 / 56.7 316.6 / 255.4 / 325.2	100.0 / 100.0 / 100.0 100.0 / 75.4 / 73.8
G: Synth. Noise Stress (Uncalibrated, seed=202603)	A B	180,049.49 180,047.57	137,929.45 137,927.48	−2,498.98 −2,430.69	779,656.47 779,316.47	131,666 / 159,504 / 156,068 131,666 / 159,444 / 156,122	100.0 / 0.1 / 0.0 100.0 / 0.0 / 0.0

5.2 Channel 2 Performance Summary: Crowd Force $F_{L,4}$ (Units: N)

Test Case & Archive Provenance	Br.	RMSE (N)	MAE (N)	Bias (N)	Max |Err| (N)	Rev MAE Pre / Dur / Post	CovFric / CovLag / CovRaw (%)
A: Zero Load (`M2_null.mat`, *N=751*)	A B	4.36 472.84	3.91 471.62	−0.77 +89.36	6.19 479.93	0.59 / 0.14 / 0.58 479.4 / 359.5 / 471.0	100.0 / 100.0 / 100.0 100.0 / 100.0 / 100.0
B1: Constant Rev. (*−50 kN, N=751*)	A B	4.36 472.84	3.91 471.62	−0.77 +89.36	6.19 479.93	0.59 / 0.14 / 0.58 479.4 / 359.5 / 471.0	100.0 / 100.0 / 100.0 100.0 / 100.0 / 100.0
B2: Constant Unidir. (*$t \in [15.5, 24.5]$ s, N=271*)	A B	4.60 475.69	4.31 475.69	+4.31 −475.69	6.19 479.01	0.0 / 0.0 / 0.80 0.0 / 0.0 / 478.9	100.0 / 100.0 / 100.0 100.0 / 100.0 / 100.0
C: Sinusoidal (`M2_100kN`, *86.5 kN* pk)	A B	1,766.46 1,728.73	1,557.23 1,546.89	−177.36 −87.24	2,951.72 3,091.48	629 / 1,354 / 1,413 1,014 / 1,425 / 1,273	100.0 / 100.0 / 13.7 100.0 / 100.0 / 12.9
D: Step Load (*60 kN* at *t=12 s, N=751*)	A B	1,862.51 1,891.42	123.75 580.35	−120.61 −30.48	43,797.58 43,317.58	0.59 / 0.14 / 0.58 479.4 / 359.5 / 471.0	100.0 / 100.0 / 99.1 100.0 / 100.0 / 99.2

Test Case & Archive Provenance	Br.	RMSE (N)	MAE (N)	Bias (N)	Max |Err| (N)	Rev MAE Pre / Dur / Post	CovFric / CovLag / CovRaw (%)
E: Time-Varying Digging ($65\ kN$ pk)	A B	635.48 905.84	458.57 754.29	−6.04 +84.08	1,574.27 2,054.27	471.3 / 474.5 / 387.2 470.6 / 595.8 / 805.6	100.0 / 100.0 / 60.9 100.0 / 100.0 / 39.7
F: Reversal-Focused ($2.0\ kN$, $0.18\ Hz$)	A B	79.64 480.19	71.67 472.49	+5.40 +95.52	119.07 590.18	62.8 / 59.5 / 60.6 417.2 / 356.7 / 485.3	100.0 / 100.0 / 100.0 100.0 / 100.0 / 52.5
G: Synth. Noise Stress (Uncalibrated, `seed=202603`)	A B	4,735.41 4,717.40	3,812.55 3,802.98	−164.74 −76.43	15,033.50 14,553.50	3,821 / 4,049 / 3,586 3,768 / 4,025 / 3,646	100.0 / 9.2 / 8.9 100.0 / 8.7 / 8.3

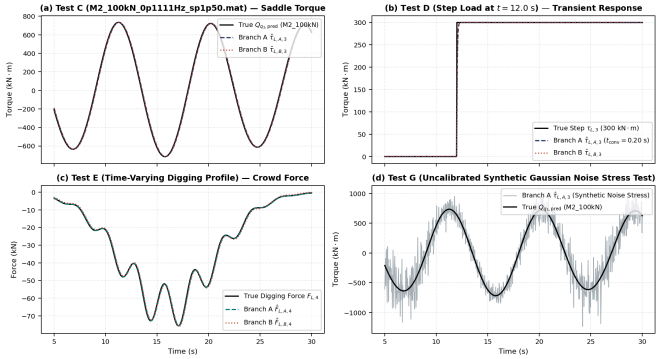


Figure 4: Time-domain tracking overlays across Test Cases C (M2_100kN_0p1111Hz_sp1p50.mat), D (Step at t=12 s), E (Digging Profile), and G (Uncalibrated Synthetic Gaussian Noise Stress Test).

5.3 Reversal-Focused Behavior, Unidirectional Bias (B1 vs. B2) & Coverage Decomposition

- **Reversing (B1) vs. Unidirectional (B2) Constant-Load Bias:** In `M2_null.mat`, the joint velocities reverse sign at $t = 5.10, 15.10, 25.10\ s$. Over the full $[5.0, 30.0]\ s$ window (`B1`), $15\ s$ of positive velocity and $10\ s$ of negative velocity partially cancel Branch B's unmodeled friction error, yielding a mean bias of $+51.23\ N{\cdot}m$ and $+89.36\ N$ despite an MAE of $297.62\ N{\cdot}m$ and $471.62\ N$. Restricting evaluation to the strictly unidirectional extension window $t \in [15.5, 24.5]\ s$ (`B2`, where $\dot{q}_3 > 0, \dot{d}_4 > 0$) exposes Branch B's full unidirectional bias: $\mathbf{-304.10\ N{\cdot}m}$ on q_3 ($-340.00\ N{\cdot}m$ friction + $35.90\ N{\cdot}m$ FOH inertial chord offset) and $\mathbf{-475.69\ N}$ on d_4 ($-480.00\ N$ friction + $4.31\ N$ FOH chord offset), vs. $+35.90\ N{\cdot}m$ and $+4.31\ N$ in Branch A.
- **Why `CovFric = 100%` While `CovRaw` Drops Under Dynamic Loads:** Across all eight test cases, `CovFricPct` ($|\hat{\tau}_{L,A,i} - \hat{\tau}_{L,B,i}| \le f_{ci}$) is identically 100.0%, confirming Young's inequality. For lag-compensated errors (`CovLagPct`), coverage is 100.0% on d_4 and 68.3%–75.4% on q_3 (where the $\le 89.24\ N{\cdot}m$ FOH inertial chord error briefly pushes $|\hat{\tau}_{L,B,3} - \tau_{L,filt,3}|$ to $349.65\ N{\cdot}m > 340\ N{\cdot}m$). However,

uncompensated raw coverage (`CovRawPct`) drops to *0.8%* (q_3) and *12.9%* (d_4) in Case C because $\pm f_{ci}$ does not bound the *50 ms* dynamic tracking lag $\tau_L - \tau_{L,filt}$.

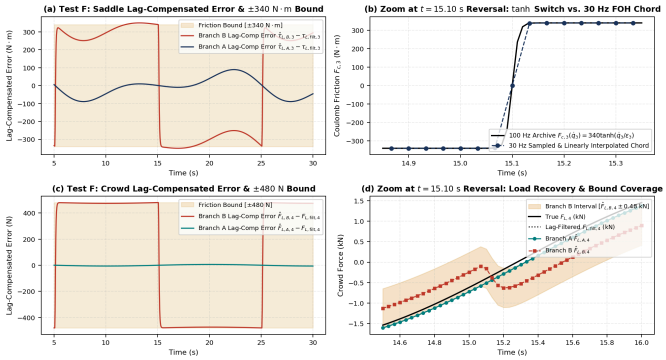


Figure 5: Test Case F (Reversal-Focused Low-Load Regime on M2_null.mat): (a, c) lag-compensated errors against the ±340 N·m and ±480 N bounds; (b) zoom at the t = 15.10 s reversal comparing the 100 Hz archive tanh switch against the 30 Hz FOH chord; (d) crowd load recovery and Branch B interval coverage.

---

6. Audited Task-Space Jacobian Factorization & Equivalent Tip-Force Critique

In recover_load.m (lines 16–19 and 182–190), an external planar force $F = [F_x, F_y]^T$ acts at handle station $s(d_4) = d_4 + k_{station}$, where $k_{bail} = +1.50\ m$ ($s_{bail} = d_4 + 1.50\ m$, observable bail arm) and $k_{COG} = c = -1.32\ m$ ($s_{COG} = r = d_4 - 1.32\ m$, negative-control COG arm). With forward kinematics $x_s(q) = [s(d_4)\ cos(q_3),\ s(d_4)\ sin(q_3)]^T$ and coordinate ordering $q = [q_3, d_4]^T$ (revolute angle first, prismatic extension second), the 2×2 task-space Jacobian is:

$$J_s(q) = \partial x_s / \partial q = [\ -s(d_4)\sin(q_3),\ \ \cos(q_3)\ ;\ \ s(d_4)\cos(q_3),\ \ \sin(q_3)\ ] = [\ \hat{e}_\theta(q_3),\ \ \hat{e}_r(q_3)\ ]\ \mathrm{diag}(s(d_4), 1) = R(q_3)\ \Pi\ \mathrm{diag}(s(d_4), 1),$$

where $R(q_3) = [cos(q_3), -sin(q_3); sin(q_3), cos(q_3)] \in SO(2)$ is the planar rotation matrix ($det\ R = +1$) and $\Pi = [0, 1; 1, 0]$ ($det\ \Pi = -1$) is the column-swap permutation matrix that places the transverse unit vector $\hat{e}_\theta = [-sin(q_3), cos(q_3)]^T$ in column 1 and the radial unit vector $\hat{e}_r = [cos(q_3), sin(q_3)]^T$ in column 2. Because $Q(q_3) = R(q_3)\Pi$ is symmetric and orthogonal ($Q(q_3)^T = Q(q_3) = Q(q_3)^{-1}$, $det\ Q = -1$), the determinant, virtual-work torque mapping $Q = J_s(q)^T F$, inverse transpose $J_s(q)^{-T}$, and equivalent tip-force mapping $F_{equiv} = J_s(q)^{-T}\ \tau$ are:

$$\det(J_s(q)) = -s(d_4), \quad Q = J_s(q)^T F = [\ s(d_4)(F_y\cos(q_3) - F_x\sin(q_3))\ ;\ \ F_x\cos(q_3) + F_y\sin(q_3)\ ],$$

$$J_s(q)^{-T} = R(q_3)\ \Pi\ \mathrm{diag}(1 / s(d_4),\ \ 1) = [\ -\sin(q_3)/s(d_4),\ \ \cos(q_3)\ ;\ \ \cos(q_3)/s(d_4),\ \ \sin(q_3)\ ],$$

$$F_{equiv} = J_s(q)^{-T}\ [\tau_3\ ;\ F_4] = (\tau_3 / s(d_4))\ \hat{e}_\theta(q_3) + F_4\ \hat{e}_r(q_3) = R(q_3)\ [\ F_4\ ;\ \ \tau_3 / s(d_4)\ ], \quad \kappa_2(J_s(q)) = \max(|s(d_4)|, 1) / \min(|s(d_4)|, 1).$$

Evaluating these exact expressions on M2_100kN_0p1111Hz_sp1p50.mat ($d_4 \in [6.7505, 10.7495]\ m$, Figure 6) reproduces `R.sig.Qq3_pred` and `R.sig.Qd4_pred` to $1.19{\times}10^{-4}\ N{\cdot}m$ and recovers `R.sig.Fx`, `R.sig.Fy` via $J_s^{-T}\ Q_{pred}$ to $2.91{\times}10^{-11}\ N$. Across the archive stroke, $s_{COG} \in [5.43, 9.43]\ m$ ($\kappa_2(J_{COG}) \in [5.43, 9.43]$) and $s_{bail} \in [8.25, 12.25]\ m$ ($\kappa_2(J_{bail}) \in [8.25, 12.25]$), with equivalent bail tip-force Coulomb floor $||J_{bail}^{-T}\ F_c||_2 = \sqrt{((340/s_{bail})^2 + 480^2)} \in [480.80, 481.77]\ N$ (*0.481%–0.482%* of the *100 kN* bail load).

Because $\kappa_2(J_s)$ mixes rotational (m/rad) and translational (m/m) columns and $||F_{equiv}||_2$ masks single-channel zero crossings, the coordinate-wise ratios $340 / |\tau_{L,3}|$ and $480 / |F_{L,4}|$ remain the primary M3 metrics.

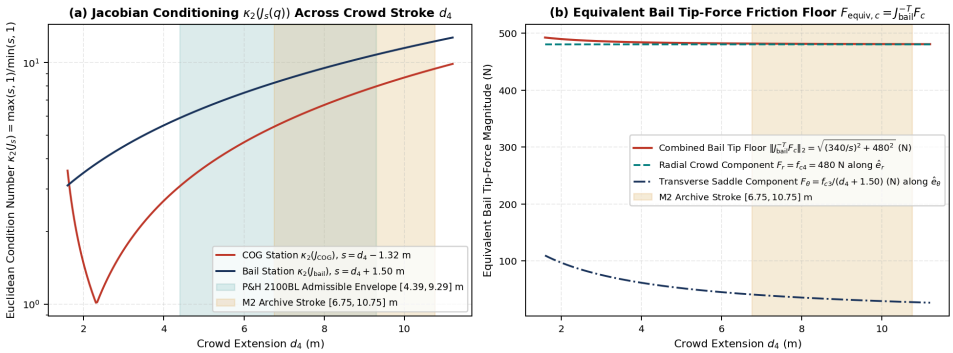


Figure 6: Audited task-space Jacobian condition number $\kappa_2(J_s)$ across d_4 for J_{COG} ($s = d_4 - 1.32$ m) vs. J_{bail} ($s = d_4 + 1.50$ m) (left), and equivalent bail tip-force Coulomb floor $||J_{bail}^{-T} F_c||_2 \in [480.80, 481.77]$ N across the M2 archive stroke [6.75, 10.75] m (right).

---

7. Explicit Limitations, Assumptions & Code-to-Report Audit Lock

In compliance with EXPERIMENT_DEFENSIBILITY_PROTOCOL.md and CLAIMS_INTEGRITY_PROTOCOL.md, the following five audit locks govern all Milestone M3 deliverables:

1. **Strict Archive Ingestion & Zero Silent Fallbacks:** `m3_observer_suite.m` (`v2.0.0-PRELIMINARY`) loads M2_null.mat and M2_100kN_0p1111Hz_sp1p50.mat directly and aborts with a diagnostic error if required time-series variables are missing (verified on stage1_results.mat). While Test A and Test C use the unmodified logged loads from `M2_null.mat` and `M2_100kN_0p1111Hz_sp1p50.mat`, Tests B1, B2, D, E, and F superimpose their analytical load waveforms onto the `M2_null.mat` Simscape trajectory because dedicated Simulink runs for those four waveforms have not yet been generated.
2. **Linear Inter-Sample Interpolation Qualification:** First-Order Hold interpolation of $p(t)$ and $h_A(t)$ between $30\ Hz$ updates is a numerical smoothness assumption, not a high-rate measurement. On `M2_null.mat`, the $33.33\ ms$ FOH chord across curved inertial/gravity trajectories introduces a baseline floor of $53.47\ N{\cdot}m$ RMS on q_3 and $4.36\ N$ RMS on d_4 at $k = 20\ s^{-1}$.
3. **Separation of Friction Bounds from Dynamic Observer Lag:** Branch B's interval $[\hat{\tau}_{L,B} - f_c,\ \hat{\tau}_{L,B} + f_c]$ strictly bounds the filtered friction discrepancy $|\hat{\tau}_{L,A} - \hat{\tau}_{L,B}| \le f_c$ (100.0% coverage), but never bounds the first-order dynamic tracking lag $\tau_L - \tau_{L,filt} \approx \dot{\tau}_L / k_i$ ($16.69\ kN{\cdot}m$ RMS on q_3 and $1.77\ kN$ RMS on d_4 in Case C).
4. **Uncalibrated Synthetic Noise Stress Test (Case G):** Test Case G applies uncalibrated zero-mean Gaussian perturbations ($\sigma_{\dot{q}3} = 3\ mrad/s$, $\sigma_{\dot{d}4} = 5\ mm/s$, portable seed `202603`) as a sensitivity stress test. Because $D_{11}(d_4)$ reaches $5.12{\times}10^6\ kg{\cdot}m^2$ at $d_4 = 10.75\ m$, $k_3\, D_{11}\, \sigma_{\dot{q}3}$ amplifies $3\ mrad/s$ velocity noise into $180.05\ kN{\cdot}m$ RMS torque noise—demonstrating why hardware encoder/tachometer specifications must be established before M4/M5 deployment.
5. **PRELIMINARY Status Pending Frozen `MS02P05` Signals:** All results in this document remain **PRELIMINARY** until `05_configuration_sweep` (`MS02P05`) is frozen in Simulink and ingested via `m3_observer_suite('M2_null.mat', 'MS02P05_archive.mat')`.